

# Science

**Year 9**

Cambridge Lower Secondary  
Student Manual

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Prime Books • Science Year 9 • Student Book

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British English edition.

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| 1.3     | The carbon cycle          |
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## INTRODUCTION

# How this book works

Almost every atom of carbon in your body was once carbon dioxide in the air, and photosynthesis put it there. In this first unit you will follow carbon from the atmosphere into living things and back again, and see why adding ancient carbon to that cycle is changing the climate.

## What makes this book different

- **Everything is original.** Every sentence, example and diagram was made for this book.

- **Every claim is checked.** Facts come from real, cited sources; every numerical answer was executed by computer before publication.

- **Thinking is taught explicitly.** Dedicated panels show you how specialists in this subject actually reason.

## A note to teachers and parents

This book follows the spirit of an international lower-secondary programme, interpreted afresh for Prime School. Each topic opens with clear outcomes and closes with self-assessment, so progress is visible to pupil and teacher alike.

## HOW TO USE THIS BOOK

# Reading the page

Each section follows the same rhythm, so you always know where you are.

| PANEL                         | WHAT IT DOES                                 |
|-------------------------------|----------------------------------------------|
| <b>Learning outcomes</b>      | What you will be able to do by the end.      |
| <b>Getting started</b>        | A warm-up that wakes up what you know.       |
| <b>Key words</b>              | The vocabulary of the topic, defined.        |
| <b>Worked example</b>         | A model solution, with the reasoning shown.  |
| <b>Questions</b>              | Practice, from straightforward to demanding. |
| <b>Investigation</b>          | A hands-on experiment with equipment.        |
| <b>Safety</b>                 | How to carry out the work safely.            |
| <b>Think like a scientist</b> | A task that builds scientific thinking.      |
| <b>Summary checklist</b>      | 'I can' statements to check your progress.   |

## How to work

Read with a pencil in your hand. When the book asks a question, try to answer it before reading on. Do the work in your notebook, and check your reasoning, not just your answer.

## GETTING SET UP

# Before Unit 1

You need a pencil, a ruler, a notebook and your curiosity. Anything else this unit requires is listed where it is needed.

## Your toolkit

- A notebook for working, planning and recording, not just for final answers.
- The habit of checking: recalculate, reread, and ask whether the answer is sensible.
- The confidence to say "I am not sure yet", which is where all real learning starts.

## A quick self-check

- 1 What do you already know about this topic?
- 2 Which part of it do you expect to find hardest, and why?
- 3 How will you know when you have genuinely understood something, rather than just recognised it?

Unit 1 begins now. Work carefully, check honestly, and ask good questions.



# 1

## Photosynthesis and the carbon cycle

How life captures sunlight, and why the climate is changing

### © IN THIS UNIT, YOU WILL

- ✓ Explain photosynthesis and write its equation.
- ✓ Describe how leaves are adapted to their job.
- ✓ Investigate the factors that limit photosynthesis.
- ✓ Trace carbon through the carbon cycle.
- ✓ Explain the evidence for climate change.

## ◆ GETTING STARTED

Almost every atom of carbon in your body was once carbon dioxide in the air. In pairs, discuss:

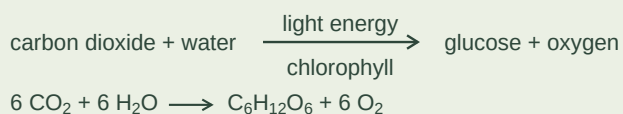
- how do you think carbon got from the air into your body?
- what living process could possibly have started that journey?
- where might those carbon atoms go after you have finished with them?

## UNIT 1 • TOPIC 1.1

# Photosynthesis

## Learn

**Photosynthesis** is the process that keeps almost every living thing alive. In it, plants use light energy to build glucose from carbon dioxide and water. It is the only significant way that energy enters the living world.



Photosynthesis happens inside **chloroplasts**, which contain the green pigment **chlorophyll**. Chlorophyll absorbs light, mostly red and blue light, and reflects green, which is why leaves look green to you.

Notice the elegance of it: respiration, which you met in Year 8, is almost exactly the reverse. Photosynthesis stores energy; respiration releases it.



Watch: Photosynthesis



Figure 1.1 A chloroplast, where light energy is captured.

### Chloroplasts and chlorophyll

Chloroplasts are small green organelles in plant cells. A single leaf cell may contain fifty or more. Each has two membranes and is filled with fluid called the stroma.

Suspended in it are stacks of flattened discs called grana; each disc is a thylakoid membrane, and this is where the chlorophyll sits.

Chlorophyll is a pigment: it strongly absorbs red and blue light and uses their energy for photosynthesis, but it cannot use green light, so green is reflected into your eyes.

That is why plants look green.

### More about photosynthesis

The glucose made by photosynthesis has three main uses. It is used in respiration, releasing energy for growth. It is converted into starch for storage. And it is turned into cellulose for cell walls, and into proteins for growth, when combined with minerals from the soil.

Photosynthesis also makes the oxygen you are breathing right now. Almost all the oxygen in the air came from ocean plankton.

#### QUESTIONS

- 1 Write the word equation and the balanced symbol equation for photosynthesis.
- 2 Where in a plant cell does photosynthesis happen?
- 3 Why do leaves appear green?
- 4 Compare photosynthesis with respiration. Give one similarity and two differences.
- 5 A plant is kept in complete darkness for a week. Explain what happens to it and why.



## ■ KEY WORDS

- **photosynthesis**: making glucose from carbon dioxide and water using light.
- **chlorophyll**: the green pigment that absorbs light energy.
- **chloroplast**: the part of a plant cell where photosynthesis happens.
- **stoma**: a tiny pore in a leaf that lets gases in and out. (Plural: stomata.)
- **limiting factor**: the factor that is currently restricting the rate.

## UNIT 1 • TOPIC 1.2

# Factors limiting photosynthesis

## Learn

A leaf is a solar panel with plumbing. Every feature has a purpose.

| FEATURE                               | HOW IT HELPS                               |
|---------------------------------------|--------------------------------------------|
| <b>broad and flat</b>                 | A large surface area to catch light.       |
| <b>thin</b>                           | Gases diffuse only a short distance.       |
| <b>many chloroplasts</b> near the top | Where the light is strongest.              |
| <b>stomata</b> on the underside       | Let carbon dioxide in and oxygen out.      |
| <b>veins</b>                          | Bring water in and carry glucose away.     |
| <b>waxy cuticle</b>                   | Reduces water loss without blocking light. |

Photosynthesis does not always run at full speed. A **limiting factor** is whatever is holding it back at that moment: usually **light intensity**, **carbon dioxide concentration** or **temperature**.

If you increase light intensity, the rate rises, then levels off, because something else has become the limiting factor. Temperature behaves differently: the rate rises, peaks, then falls sharply, because the enzymes that drive photosynthesis are damaged, or **denatured**, by heat.

## 🔬 INVESTIGATION: HOW LIGHT AFFECTS PHOTOSYNTHESIS

**You will need:** pondweed (*Elodea*), a test tube, a beaker of water, a lamp, a ruler, sodium hydrogen carbonate solution, a stopwatch.

### Method:

- 1 Place the pondweed in the test tube of sodium hydrogen carbonate solution, which supplies carbon dioxide.
- 2 Stand the tube in a beaker of water. This acts as a heat shield, so the lamp warms the plant as little as possible.
- 3 Place the lamp 10 cm away. Wait two minutes for the rate to settle.
- 4 Count the oxygen bubbles released in one minute. Repeat twice and take a mean.
- 5 Repeat at 20 cm, 30 cm, 40 cm and 50 cm.
- 6 Plot a graph of bubbles per minute against distance.

### ⚠ SAFETY

The lamp becomes hot. Do not touch the bulb and keep water away from the electrical fitting. Wash your hands after handling pond water.

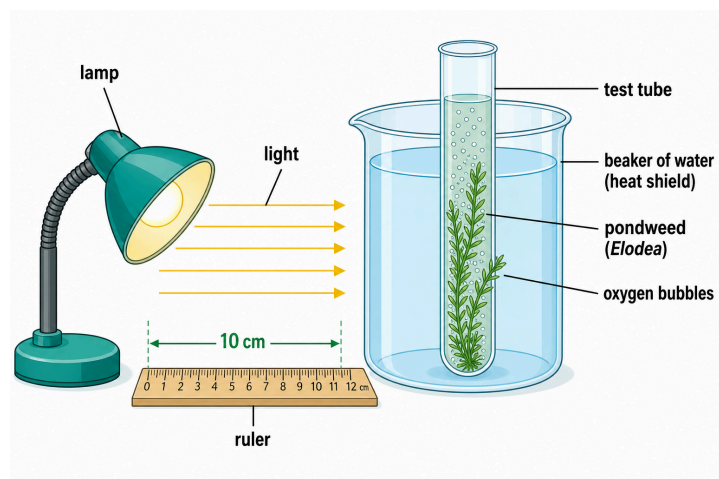


Figure 1.2 Light intensity affects photosynthesis from different distances.

- 5 In the investigation, why was a mean of three counts taken?

∴ THINK LIKE A SCIENTIST

In this investigation, why is the beaker of water essential? Which variable would become confounded without it, and how would that affect your conclusion?

• QUESTIONS

- 1 Give three adaptations of a leaf and explain how each helps photosynthesis.
- 2 What is a limiting factor?
- 3 Sketch the graph of rate against light intensity. Explain the shape of each part.
- 4 Explain why the rate of photosynthesis falls at very high temperatures, using the word denatured.
- 5 In the investigation, why was a mean of three counts taken?

# The carbon cycle

## Learn

Carbon atoms are never created or destroyed. They circulate endlessly between the air, living things, the oceans and the rocks. This circulation is the **carbon cycle**.

### Carbon goes OUT of the air by:

- **photosynthesis**, when plants build glucose from carbon dioxide

### Carbon returns TO the air by:

- **respiration**, in plants, animals and microbes
- **decomposition**, when microbes break down dead material
- **combustion**, when fuels are burned



Watch: The carbon cycle

For millions of years these processes roughly balanced. Some carbon, though, was locked away underground as **fossil fuels**: coal, oil and natural gas, formed from organisms that died long ago and did not fully decompose. That carbon was out of the cycle entirely.

When we burn fossil fuels, we return that ancient carbon to the air in a few centuries, far faster than photosynthesis and the oceans can remove it. The cycle no longer balances, and carbon dioxide accumulates.



Figure 1.2 The carbon cycle, showing the routes carbon takes.

### • QUESTIONS

- 1 Name the process that removes carbon dioxide from the air.
- 2 Name three processes that return carbon dioxide to the air.
- 3 Explain how fossil fuels formed, and why burning them changes the cycle.
- 4 A forest is cut down and burned. Give two reasons why this raises atmospheric carbon dioxide.
- 5 Explain why planting trees can help, but cannot by itself solve the problem.

### ✦ DID YOU KNOW?

Carbon is a champion recycler. A carbon atom in your breath this morning may have been in a dinosaur, a limestone cliff, a fern in a coal seam and an ocean wave, and it will still be circulating long after you have gone.

# Climate change

## Learn

Some gases in the atmosphere, called **greenhouse gases**, let sunlight through but absorb the infrared radiation the Earth emits. This keeps the planet warm enough for life, and is called the **greenhouse effect**. It is entirely natural and entirely necessary.

The problem is its strengthening. By adding carbon dioxide and methane to the atmosphere, human activity is trapping more heat than before. This is **enhanced global warming**, and the resulting shifts in weather patterns are **climate change**.

## The evidence

The measurements are real, public and easy to check. At the Mauna Loa Observatory in Hawaii, carbon dioxide has been recorded continuously since 1958.

| YEAR                          | CARBON DIOXIDE (PPM) |
|-------------------------------|----------------------|
| pre-industrial (before ~1750) | about 280            |
| 1958 (record begins)          | 315                  |
| 2000                          | 369                  |
| 2015                          | 402                  |
| 2025                          | 429.6                |
| June 2026                     | <b>431.4</b>         |

That is a rise of more than 50% above the pre-industrial value, and the rate of increase is itself increasing. Ice cores show that for the previous 800 000 years, carbon dioxide never rose above about 300 ppm.

## Consequences

- **Rising global mean temperature**, and more frequent heatwaves.
- **Melting ice sheets and glaciers**, and **rising sea level**.
- **Ocean acidification**, as seawater absorbs carbon dioxide.
- **Shifting rainfall**, bringing drought to some regions and floods to others.

Portugal is directly affected. Longer, hotter summers raise wildfire risk in the interior, and the Algarve faces increasing pressure on its water supply.



### ∴ THINK LIKE A SCIENTIST

Scientists do not rely on a single measurement. The Mauna Loa record is supported by ice cores, satellite data, ocean buoys and thousands of independent stations, all agreeing. Explain why this **convergence of independent evidence** makes a conclusion far stronger than any one dataset could.

### • QUESTIONS

- 1 Explain the natural greenhouse effect, and why life needs it.
- 2 Using the table, calculate the rise in carbon dioxide from 1958 to June 2026.
- 3 What percentage increase is 431.4 ppm compared with the pre-industrial 280 ppm?  
Show your working.
- 4 Give three consequences of enhanced global warming.
- 5 Describe two effects of climate change that matter specifically in Portugal.

### ☒ SUMMARY CHECKLIST

- ✓ I can write the equation for photosynthesis and explain it.
- ✓ I can explain how leaves are adapted to photosynthesis.
- ✓ I can identify and explain limiting factors.
- ✓ I can trace carbon through the carbon cycle.
- ✓ I can describe the evidence for climate change and evaluate it.

## UNIT 1 • FINAL PROJECT

# A carbon audit of your school

★ FINAL PROJECT

## A carbon audit of your school

Work in groups to investigate where carbon dioxide enters and leaves your school grounds, then propose one realistic change that would reduce net emissions. Present your findings as a scientific poster with data.

### What to hand in

- A labelled map of your school showing carbon sources and carbon sinks.
- At least three real measurements or counts you made yourselves.
- One realistic proposal, with an estimate of its effect and your reasoning.
- A sources list, with links, for any external data you used.

### How it will be judged

| Criterion | Weight | |---|---| | Scientific accuracy of the carbon analysis | 35% | | Quality and honesty of your own data | 30% | | Realism of the proposal and its justification | 25% | | Presentation and referencing | 10% |

## UNIT 1 • CHECK YOUR PROGRESS

- 1 Write the balanced symbol equation for photosynthesis. [2]
- 2 Name the pigment that absorbs light, and the structure that contains it. [2]
- 3 Give three leaf adaptations and explain each. [3]
- 4 Explain what is meant by a limiting factor, with an example. [2]
- 5 Name one process that removes carbon dioxide from the air and three that return it. [4]
- 6 Explain why burning fossil fuels unbalances the carbon cycle. [2]
- 7 The Mauna Loa record shows 315 ppm in 1958 and 431.4 ppm in June 2026.  
a. Calculate the increase. b. Calculate the mean annual rise. [2]
- 8 Explain why the greenhouse effect is natural but enhanced global warming is a problem. [2]

## UNIT 1 • EVALUATION

Be honest. Colour the circle that matches how you feel about each outcome: green means confident, amber means nearly there, red means needs more work.

| I CAN ...                                | GREEN                 | AMBER                 | RED                   |
|------------------------------------------|-----------------------|-----------------------|-----------------------|
| explain photosynthesis and its equation  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| explain leaf adaptations                 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| investigate limiting factors             | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| trace carbon through the carbon cycle    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| evaluate the evidence for climate change | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

## UNIT 1 • WHAT CAN YOU DO?

### ✓ WHAT CAN YOU DO?

- |                                                                               |                                                                                 |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <input type="checkbox"/> Write and explain the equation for photosynthesis.   | <input type="checkbox"/> Trace carbon through the carbon cycle.                 |
| <input type="checkbox"/> Explain how a leaf is adapted to its function.       | <input type="checkbox"/> Describe and evaluate the evidence for climate change. |
| <input type="checkbox"/> Identify limiting factors and interpret rate graphs. |                                                                                 |

## ■ UNIT 1 ROUND-UP

photosynthesis · chlorophyll · chloroplast · stoma · limiting factor · denatured · carbon cycle · decomposition · combustion · fossil fuel · greenhouse gas · greenhouse effect · climate change · ocean acidification

## GLOSSARY

### Unit 1

**photosynthesis:** making glucose from carbon dioxide and water using light.

**chlorophyll:** the green pigment that absorbs light energy.

**chloroplast:** the part of a plant cell where photosynthesis happens.

**stoma:** a tiny pore in a leaf that lets gases in and out.

**limiting factor:** the factor currently restricting the rate.

**denatured:** changed in shape by heat so it no longer works.

**carbon cycle:** the circulation of carbon between air, life, oceans and rocks.

**greenhouse effect:** the trapping of infrared radiation by atmospheric gases.

**ocean acidification:** the fall in ocean pH as seawater absorbs carbon dioxide.

## SOURCES & REFERENCES

### Where everything came from

All text and diagrams are original to this edition. The factual claims can be checked here. Links were live as of July 2026.

## § UNIT 1 • PHOTOSYNTHESIS AND THE CARBON CYCLE

- 1 **Atmospheric carbon dioxide at Mauna Loa. The monthly mean for June 2026 was 431.44 ppm; the pre-industrial value was about 280 ppm.**  
<https://gml.noaa.gov/ccgg/trends/>
- 2 **The 2025 seasonal peak. Atmospheric carbon dioxide peaked at 430.5 ppm in May 2025.**  
<https://gml.noaa.gov/ccgg/trends/weekly.html>
- 3 **Ice-core evidence. For the previous 800 000 years, carbon dioxide never exceeded about 300 ppm.**  
<https://www.climate.gov/news-features/understanding-climate/climate-change-atmospheric-carbon-dioxide>
- 4 **Photosynthesis. The light-driven reaction that builds glucose from carbon dioxide and water.**  
<https://en.wikipedia.org/wiki/Photosynthesis>

# Follow the carbon.

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This Year 9 Student Book connects a leaf to the whole planet. You will write the equation for photosynthesis, investigate what limits it, trace carbon through its cycle, and evaluate the real evidence for climate change.

Written for Prime School pupils, in clear British English, with original diagrams and real, cited data.

PHOTOSYNTHESIS

CARBON CYCLE

CLIMATE

EVIDENCE

INVESTIGATION